

analysis capability, which examines RTL-level switching activity to predict power, timing, or electromagnetic leakage before fabrication. Traditionally, such vulnerabilities are identified post-silicon, when mitigation is costly or impractical. Validated against FPGA-level measurements, this approach enables corrective action during the design phase.

Industry-academia collaboration has played a significant role in shaping the platform. Engagement with the Shakti Processors research groups at IIT Madras provided exposure to real-world RISC-V implementation challenges, early validation platforms, and ecosystem credibility. Over time, the company has evolved from offering standalone verification IP blocks to delivering an integrated platform environment, marking a shift towards platform-as-a-service delivery.

Operating with a compact team of around five engineers, the firm demonstrates high-leverage engineering. AI copilots and orchestration systems support code development, regression management, and coverage analysis, enabling output comparable to much larger teams. The company follows a subscription-based revenue model, with both cloud and on-premise deployment options, serving RISC-V startups, academic institutions, and established semiconductor companies,



Lavanya Jagadeeswaran, CEO, Vyoma Systems, India

including international clients.

As a RISC-V quality and security solutions provider, the company does not manufacture chips or operate fabrication plants. Instead, it supports fabless semiconductor companies and design houses through an asset-light strategy focused on pre-silicon design verification, emulation, software bring-up, side-channel analysis, and domain expertise. Growth has been largely bootstrapped, supplemented by funding from IITM Pravartak Technologies Foundation, IIT Madras Incubation Cell, and Tamil Nadu's Startup TN TANSEED grant, with revenues driven primarily by customer subscriptions rather than venture capital.

Investments have prioritised advanced compute infrastructure, FPGA prototyping platforms, security tooling, and specialised talent rather than physical manufacturing assets. Given the trust-sensitive nature of semiconductor verification, building credibility remains a central challenge, particularly when handling proprietary IP and integrating AI-driven workflows into quality-critical processes.

Looking ahead, the company aims to deepen AI integration across the verification lifecycle, expand partnerships within India, and enter global RISC-V markets in the United States and Europe. Future growth will depend on scaling engineering talent, strengthening ecosystem collaborations, and onboarding domain-aware channel partners with established semiconductor networks. Continued academic partnerships in hardware security, formal verification, and advanced RISC-V extensions remain central to its long-term innovation strategy.

As India accelerates its semiconductor ambitions, Vyoma Systems positions itself not just as a platform provider but as a foundational partner in enabling trustworthy silicon—ensuring that every chip design achieves uncompromising reliability and security before fabrication.

—Akanksha Sondhi Gaur

3 ARVO: THE ANSWER TO PRODUCT AUTHENTICITY CHALLENGES

A deep-tech startup using AI-powered physical patterns to combat counterfeiting and improve traceability.

Founded in 2023 by Duke Banerjee and Sumit Goswami, OneArvo Ventures Private Limited, operating under the brand name ARVO, is focused on addressing key challenges faced by brands, including counterfeiting.

Before founding the startup,

both founders worked together at a private equity firm, where exposure to multiple portfolio companies revealed recurring issues around counterfeiting, poor traceability, and weak verification across supply chains. After finding QR codes, barcodes, and holograms

inadequate, they set out to build a more robust, future-ready alternative. The company is registered in Delhi, with its operational headquarters and lab facilities located in Noida.

The startup's solution is based on a non-clonable physical pattern